International AS · Edexcel · Chemistry

8 mins 7 questions

Multiple Choice Questions

Group 7

Trends of the Group 7 Elements / Halogen Displacement Reactions / Redox Reactions of the Halogens / Other Reactions of the Halogens

Easy (2 questions)	/2
Medium (5 questions)	/6
Total Marks	**/8**

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Easy Questions

1 Which statement about the Group 7 elements chlorine, bromine and iodine is **not** correct?

A. boiling temperature increases down the group

B. reactivity increases down the group

C. first ionisation energy decreases down the group

D. electronegativity decreases down the group

(1 mark)

2 Which silver halides are soluble in **concentrated** aqueous ammonia?

A. AgBr and AgI

B. AgCl and AgI

C. AgCl and AgBr

D. AgCl only

(1 mark)

Medium Questions

1 (a) This question is about the reaction of potassium iodide with concentrated sulfuric acid.

Which of these would **not** be seen?

A. misty fumes

B. black solid

C. yellow solid

D. dense white smoke

(1 mark)

(b) One of the reaction products is the gas H_2S. What is the **change** in oxidation number of sulfur as H_2S forms?

A. −8

B. −6

C. −2

D. +6

(1 mark)

2 Which of these **increases** as Group 7 is descended?

A. oxidising ability of the molecular halogens

B. reducing ability of the halide ions

C. electrostatic attraction between the nucleus and outer shell of electrons

D. electronegativity of the halogen atoms

(1 mark)

3 When iodine is dissolved in a non-polar organic solvent, the solution formed is

A. purple

B. orange

C. colourless

D. brown

(1 mark)

4 Four tests used to identify ions are shown:

1 flame test
2 addition of acidified barium nitrate solution
3 addition of acidified silver nitrate solution
4 addition of sodium hydroxide solution, then testing any gas with indicator paper

Which tests could be used to positively identify the ions in ammonium chloride?

A. 1 and 2

B. 1 and 3

C. 2 and 4

D. 3 and 4

(1 mark)

5 Solid potassium bromide reacts with concentrated sulfuric acid.

Which of these substances does **not** form?

A. bromine

B. hydrogen bromide

C. hydrogen sulfide

D. sulfur dioxide

(1 mark)